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Inventor(s)	SATTARZADEH HASHEMI; Ata et al.

LOW COMPLEXITY PRECODER DESIGN FOR MULTI-USER MULTI-INPUT MULTI-OUTPUT ORTHOGONAL TIME AND FREQUENCY SPACE NETWORKS

Abstract

A device may apply a precoder to input information symbols to generate delay-Doppler (DD) domain symbols, and may utilize an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal. The device may utilize a Heisenberg transform to convert the TF domain signal to a time domain signal, and may add a cyclic prefix to the time domain signal to generate a modified time domain signal. The device may transmit the modified time domain signal to at least one user equipment.

Inventors: **SATTARZADEH HASHEMI; Ata (Guildford, GB), DIZDAR; Onur (London, GB), BATTULA; Venkateswara (Bangalore, IN), WANG; Stephen (London, GB)**

Applicant: **VIAVI Solutions Inc.** (Chandler, AZ)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This Patent Application claims priority to Indian patent application Ser. No. 20/241,1013648, filed on Feb. 26, 2024, and entitled "LOW COMPLEXITY PRECODER DESIGN FOR MULTI-USER MULTI-INPUT MULTI-OUTPUT ORTHOGONAL TIME AND FREQUENCY SPACE NETWORKS." The disclosure of the prior Application is considered part of and is incorporated by reference into this Patent Application.

BACKGROUND

[0002] New waveform candidates, such as orthogonal time and frequency space (OTFS) waveforms, may be used to address current shortcomings of orthogonal frequency division multiplexing (OFDM) waveforms.

SUMMARY

[0003] Some implementations described herein relate to a method. The method may include applying a precoder to input information symbols to generate delay-Doppler (DD) domain symbols, and utilizing an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal. The method may include utilizing a Heisenberg transform to convert the TF domain signal to a time domain signal, and adding a cyclic prefix to the time domain signal to generate a modified time domain signal. The method may include transmitting the modified time domain signal to at least one user equipment.

[0004] Some implementations described herein relate to a device. The device may include one or more memories and one or more processors coupled to the one or more memories. The one or more processors may be configured to apply a precoder to input information symbols to generate delay-Doppler (DD) domain symbols, and utilize an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal. The one or more processors may be configured to utilize a Heisenberg transform to convert the TF domain signal to a time domain signal, and add a cyclic prefix to the time domain signal to generate a modified time domain signal. The one or more processors may be configured to transmit the modified time domain signal to at least one user equipment to cause the at least one user equipment to convert the modified time domain signal to the TF domain signal and to convert the TF domain signal to the DD domain symbols.

[0005] Some implementations described herein relate to a non-transitory computer-readable medium that stores a set of instructions. The set of instructions, when executed by one or more processors of a device, may cause the device to apply a precoder to input information symbols to generate delay-Doppler (DD) domain symbols, and utilize an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal. The set of instructions, when executed by one or more processors of the device, may cause the device to utilize a Heisenberg transform to convert the TF domain signal to a time domain signal, wherein the inverse symplectic finite Fourier transform and the Heisenberg transform are provided in an orthogonal time and frequency space modulator of the device. The set of instructions, when executed by one or more processors of the device, may cause the device to add a cyclic prefix to the time domain signal to generate a modified time domain signal, and transmit the modified time domain signal to at least one user equipment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIGS. **1**A-**1**H are diagrams of an example associated with applying zero-forcing (ZF) and minimum mean square error (MMSE) or any other linear precoder to multi-user multi-input multi-output (MU-MIMO) OTFS networks.

[0007] FIG. **2** is a diagram of an example environment in which systems and/or methods described herein may be implemented.

[0008] FIG. **3** is a diagram of example components of one or more devices of FIG. **2**.

[0009] FIG. **4** is a flowchart of an example process for applying ZF and MMSE to MU-MIMO OTFS networks.

DETAILED DESCRIPTION

[0010] The following detailed description of example implementations refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

[0011] OFDM is a widely used waveform in wireless communication systems due to high spectral efficiency, robustness to multipath fading, and simple implementation of channel equalization. However, OFDM has several drawbacks, such as a high peak-to-average power ratio (PAPR), where the peak power can be much higher than the average power. This may require use of highly linear power amplifiers, which are less power-efficient and can lead to increased power consumption and reduced battery life in mobile devices. OFDM is sensitive to frequency offset and Doppler shift, which can cause inter-carrier interference (ICI). OFDM utilizes a cyclic prefix to combat inter-symbol interference (ISI) due to multipath propagation. The cyclic prefix may reduce the overall spectral efficiency of an OFDM waveform.

[0012] Furthermore, synchronization of OFDM systems, particularly time and frequency synchronization, can be complex and challenging. OFDM also requires accurate channel estimation to equalize channel effects. OFDM includes significant out-of-band emissions, which can cause interference with adjacent channels. In environments with significant delay spread, performance of OFDM may be negatively impacted due to loss of orthogonality between subcarriers. Therefore, current techniques for generating OFDM waveforms consume computing resources (e.g., processing resources, memory resources, communication resources, and/or the like), networking resources, and/or the like associated with increasing power consumption, reducing battery life, handling inter-carrier interference, reducing spectral efficiency, generating a degraded quality signal, and/or the like.

[0013] Some implementations described herein relate to a radio access network (RAN) that applies ZF and MMSE to MU-MIMO OTFS networks. For example, the RAN may apply a precoder to input information symbols to generate delay-Doppler (DD) domain symbols, and may utilize an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal. The RAN may utilize a Heisenberg transform to convert the TF domain signal to a time domain signal, and may add a cyclic prefix to the time domain signal to generate a modified time domain signal. The RAN may transmit the modified time domain signal to at least one user equipment.

[0014] In this way, the RAN applies ZF and MMSE to MU-MIMO OTFS networks. For example, a RAN may apply ZF and/or MMSE models to beamforming techniques for MU-MIMO OTFS systems. Although the models are described in connection with MU-MIMO, the models may also be used in single user MIMO (SU-MIMO) OTFS. In a ZF precoding scheme for OTFS, a precoder for each antenna and each user may be designed to cancel out total interference to the other users. An MMSE precoder for OTFS may incorporate benefits of maximum ratio transmission (MRT) and ZF precoding schemes considering both signal and interference levels to achieve a robust performance. A transformation may be utilized to diagonalize an effective channel matrix of the system, and a precoding design may be performed accordingly. Alternatively, beamformers may be calculated and applied in the time-frequency (TF) domain rather than the DD domain. Thus, the RAN conserves computing resources, networking resources, and/or the like that would otherwise have been consumed by increasing power consumption, reducing battery life, handling inter-carrier

interference, reducing spectral efficiency, generating a degraded quality signal, and/or the like.
[0015] FIGS. **1**A-**1**H are diagrams of an example **100** associated with applying ZF and MMSE to MU-MIMO OTFS networks. As shown in FIGS. **1**A-**1**H, the example **100** may include user equipments (UEs) associated with a radio access network (RAN) and a core network. In some implementations, the UEs and the RAN may form an MU-MISO OTFS network, and the RAN may utilize ZF and MMSE precoding schemes. Further details of the UEs, the RAN, and the core network are provided elsewhere herein.
[0016] In some implementations, the following abbreviations may be used herein: additive white Gaussian noise (AWGN), broadcast channel (BC), cyclic prefix (CP), channel state information (CSI), channel state information at transmitter (CSIT), doubly block circulant (DBC), delay-Doppler (DD), degree-of-freedom (DoF), de-modulation reference signal (DMRS), discrete Zak transform (DZT), inter-carrier interference (ICI), inter-symbol interference (ISI), multi-input multi-output (MIMO), multi-input single-output (MISO), minimum mean square error (MMSE), maximum ratio transmission (MRT), multi-user multi-input multi-output (MU-MIMO), new radio (NR), symplectic finite Fourier transform (SFFT), signal-to-noise ratio (SNR), signal-to-interference-plus-noise ratio (SINR), single-user MIMO (SU-MIMO), time-frequency (TF), and zero forcing (ZF).
[0017] In some implementations, the following parameters may be used herein.
TABLE-US-00001 Parameter Definition N.sub.t Number of transmit antennas K Number of users M Number of bins in the delay domain N Number of bins in the Doppler domain Δf Subcarrier spacing in OFDM conversion T.sub.OTFS Frame length before CP P Number of resolvable paths X.sub.DD DD domain symbol matrix at transmitter X.sub.DD Vector form of DD domain symbol matrix at transmitter X.sub.TF TF domain signal s(t) Time-domain OTFS signal g.sub.tx(t) Transmitter pulse shaping filter h.sub.DD (τ, ν) Channel model in DD domain h.sub.i Complex channel coefficient of the i-th channel path τ.sub.i Delay of i-th channel path ν.sub.i Doppler shift of i-th channel path τ.sub.max Maximum delay spread ν.sub.max Maximum Doppler spread Y.sub.DD DD domain symbol matrix at receiver Y.sub.DD Vector form of DD domain symbol matrix at receiver H.sub.DD Channel matrix in DD domain H.sub.DD.sup.q, k Channel matrix in DD domain from q-th transmit antenna to user-k Ĥ.sub.DD.sup.q, k Estimated channel matrix in DD domain from q-th transmit antenna to user-k Π Permutation matrix with forward cyclic shift F.sub.N N-point discrete Fourier transform matrix Δ Diagonal matrix of phase compensation term Γ Phase compensation term F.sub.qk Precoder matrix in DD domain for q-th transmit antenna and user-k S.sub.DD DD domain symbol matrix at transmitter before precoding S.sub.TF TF domain symbol matrix at transmitter before precoding S.sub.DD.sup.k DD domain symbol vector at transmitter intended for user-k before precoding Ŝ.sub.DD.sup.k DD domain equalized symbol vector at user-k X.sub.DD.sup.q DD domain symbol vector at transmitter fed to q-th transmit antenna after precoding F.sub.qk Precoder matrix in DD domain for q-th transmit antenna and user-k Y.sub.DD.sup.k DD domain symbol vector received at user-k Z.sup.k DD domain noise matrix at user-k {circumflex over (R)}.sub.DD.sup.q, k Estimate of DD domain symbol matrix transmitted from q-th transmit antenna and received by user-k and estimated at the transmitter {circumflex over (R)}.sub.DD.sup.q, k Vector form of estimate of DD domain symbol matrix transmitted from q-th transmit antenna and received by user-k and estimated at the transmitter {circumflex over (R)}.sub.TF.sup.q, k Estimate of TF domain symbol matrix transmitted from q-th transmit antenna and received by user-k and estimated at the transmitter Ĥ.sub.TF.sup.q, k Estimate of channel matrix in DD domain from q-th transmit antenna to user-k ω.sub.m, n.sup.MMSE MMSE precoding matrix in TF domain for the TF bin [m, n] ω.sub.m, n.sup.ZF ZF precoding matrix in TF domain for the TF bin [m, n] W.sub.qk Precoding matrix in TF domain for q-th transmit antenna and user-k
[0018] FIG. **1**B is a diagram of example components of a RAN in which systems and/or methods described herein may be implemented. As shown in FIG. **1**B, a multi-input single-output (MISO)

broadcast setting may be implemented, where K single antenna users are served using N_t antennas with $N_t \geq K$. As further shown in FIG. **1**B, information symbols of different users ($u_1, u_2, \ldots, u_k$) may be provided to a precoder, where precoders may be applied to delay-domain symbols of the users. Delay-Doppler (DD) domain symbols may be modulated using an OTFS modulator. In FIG. **1**B, an inverse symplectic finite Fourier transform (ISFFT) based modulation may be utilized. In the ISFFT based modulator, each OTFS grid has M bins in the delay and N bins in the Doppler dimensions. The total bandwidth (BW) used to transmit the OTFS symbols may be $BW=M\times\Delta f$, where Δf is the subcarriers spacing used in the OFDM conversion (e.g., the Heisenberg transform) and the frame length before adding the cyclic prefix would be $T_{OTFS}=N\times T$, where $T=1/\Delta f$. Further details of the components shown in FIG. **1**B are provided elsewhere herein.

[0019] DD domain transmission schemes, such as an OTFS network, may include unique properties especially for high Doppler scenarios, and may be compatible with multiple antenna transmitters and receivers in order to increase spectral efficiency and performance of communication systems. An OTFS network may transmit data symbols in the DD domain, where a channel is quasi-static and sparse. In OTFS, transmitted symbols may be spread in the entire time and frequency domain, which enables the OTFS network to harvest full degrees of freedom of communication channels.

[0020] In some implementations, the RAN may apply beamforming techniques, such as ZF and MMSE precoding, to a MU-MIMO OTFS system. Although implementations are described in connection with MU-MIMO OTFS, the implementations may also be utilized in SU-MIMO OTFS with slight modifications. In a ZF precoding scheme for OTFS, a precoder for each antenna and each user may cancel out total interference to other users. The MMSE precoder for OTFS may include the benefits of MRT and ZF precoding schemes by considering both signal and interference levels. In some implementations, a mathematical transformation may be utilized to diagonalize an effective channel matrix of the RAN and to generate a precoding design accordingly. In some implementations, the RAN may apply the precoding in the time-frequency (TF) domain rather than the DD domain.

[0021] As shown in FIG. **1**C, and by reference number **105**, the RAN may apply a precoder to input information symbols to generate delay-Doppler (DD) domain symbols. For example, the RAN may apply one of the precoders, described below in connection with FIG. **1**D, to the input information symbols to generate the DD domain symbols. The input information symbols may be provided for different users (e.g., UEs), and may be provided to one of the precoders. The one of the precoders may process the input information symbols to generate the DD domain symbols.

[0022] As shown in FIG. **1**D, and by reference number **110**, the RAN may utilize a ZF and MMSE based precoder for MU-MIMO OTFS in a DD domain based on a doubly block circulant (DBC) property of a channel matrix. For example, the RAN may utilize a low-complexity MU-MIMO precoding scheme, and perfect channel state information (CSI) may be available at the RAN. The RAN may include N_t antenna capable of serving serve K users, where $N_t \geq K$. A precoder matrix may be defined as $F_{qk} \in$ custom-character$^{MN\times MN}$, $q \in \{1, \ldots, N_t\}$ and $k \in \{1, \ldots, K\}$. A signal fed into a qth OTFS modulator connected to a qth antenna may be calculated as:

[00001] $$X_{DD}^{q} = \sum_{k=1}^{K} F_{qk} S_{DD}^{k}, \quad (1)$$

where S_{DD}^{k} is the vectorized modulated input symbols for the kth user in the DD domain. A received DD domain signal for the kth user may be calculated as:

[00002]
$$
\begin{aligned}
Y_{\mathrm{DD}}^{k} &= \overset{N_t}{\underset{q=1}{.\mathrm{Math}.}} H_{\mathrm{DD}}^{q,k} X_{\mathrm{DD}}^{q} + Z^{k} \\
&= \overset{N_t}{\underset{q=1}{.\mathrm{Math}.}} \overset{K}{\underset{k'=1}{.\mathrm{Math}.}} H_{\mathrm{DD}}^{q,k} F_{qk'} S_{\mathrm{DD}}^{k'} + Z^{k} \\
&= \underbrace{.\mathrm{Math}._{q=1}^{N_t} H_{\mathrm{DD}}^{q,k} F_{qk} S_{\mathrm{DD}}^{k}}_{\text{Desired Signal}} + \underbrace{.\mathrm{Math}._{q=1}^{N_t} .\mathrm{Math}._{\substack{k' \in \{1, .\mathrm{Math}., K\} \\ k' \neq k}} H_{\mathrm{DD}}^{q,k} F_{\mathrm{qk}'} S_{\mathrm{DD}}^{k'}}_{\text{Interference}} + \underbrace{Z^{k}}_{\text{noise}} .
\end{aligned}
\quad (2)
$$

The RAN may obtain F.sub.qk for MU-MIMO with low-complexity methods.

[0023] The RAN may utilize the properties of a DBC matrix. A DBC matrix can be decomposed as:

[00003] $H_{\mathrm{DD}} = \ ^{-1} \Delta \Xi, \quad (3)$

where

[00004]
$$= F_M \ .\mathrm{Math}.\ F_N \quad (4)$$
$$= \mathrm{diag}(\mathrm{vec}(F_N H' F_M)) . \quad (5)$$

Where Δ is a diagonal matrix composed of eigenvalues of a channel matrix H.sub.DD and H′=vec.sub.N,M.sup.−1(h.sub.1), with h.sub.1 being the first column of the H.sub.DD.

[0024] A channel in a continuous DD domain may be modelled as:

[00005] $h_{\mathrm{DD}}(\ , \nu) = \ .\mathrm{Math}._{i=1}^{P} h_i \ (\ - \ _i) \ (\nu - \nu_i), \quad (6)$

where P is an integer number showing a number of resolvable paths and h.sub.i is a complex number showing the channel coefficient of an ith path. Maximum values for delay and Doppler of each path may be restricted to a maximum delay spread and a maximum Doppler spread of the channel. Therefore, this provides 0<τ.sub.i≤τ.sub.max and −v.sub.max≤v.sub.i≤+v.sub.max.

[0025] The relation between input and output signal in the OTFS may be provided in matrix form as follows:

[00006] $Y_{\mathrm{DD}} = H_{\mathrm{DD}} X_{\mathrm{DD}} + n . \quad (7)$

[0026] In this formulation, Y.sub.DD may be a vectorized version of a matrix Y.sub.DD, obtained by stacking up all the columns of Y.sub.DD into a single column vector, where the matrix Y.sub.DD is the matrix of received symbols in the DD domain matching the DD grid. This operation may be shown as vec(.) and a reverse operation may be shown as vec.sub.M,N.sup.−1(.). Therefore, Y.sub.DD=vec(Y.sub.DD)∈custom-character.sup.MN×1 and X.sub.DD=vec(X.sub.DD)∈custom-character.sup.MN×1. The last term n∈custom-character.sup.MN×1 is a vector the additive white Gaussian noise (AWGN) with independent and identically distributed elements with distribution custom-character(0, σ.sup.2). The elements of the channel matrix H.sub.DD∈custom-character.sup.MN×MN may be dependent on channel coefficients defined in equation (6) and may capture the quasi-periodicity properties of the OTFS. Furthermore, the effects of the choice of a cyclic prefix and pulse shaping may be reflected in this matrix.

[0027] With the ideal pulse shaping satisfying bi-orthogonality conditions, H.sub.DD may be a doubly block circulant (DBC) matrix with M circulant blocks H.sup.i each of size N×N and may be expressed as:

[00007] $$H_{\mathrm{DD}} = [\begin{matrix} H^0 & H^{M-1} & \text{.Math.} & H^2 & H^1 \\ H^1 & H^0 & \text{.Math.} & H^3 & H^2 \\ \text{.Math.} & \text{.Math.} & \ddots & \text{.Math.} & \text{.Math.} \\ H^{M-2} & H^{M-3} & \text{.Math.} & H^0 & H^{M-1} \\ H^{M-1} & H^{M-2} & \text{.Math.} & H^1 & H^0 \end{matrix}]. \quad (8)$$

This is the channel matrix model that is used for the ZF and MMSE based precoder.

[0028] The diagonalization method may provide a basis for the ZF and MMSE based precoder. The channel matrix model of equation (8), H.sub.DD, may include off diagonal elements. This may indicate that each symbol of X.sub.DD can potentially induce interference across multiple output symbols within Y.sub.DD, resulting in inter-symbol interference (ISI) and intercarrier interference (ICI). Consequently, by transforming the channel matrix into a diagonal form, the RAN may mitigate the ISI and ICI interference effectively. This transformation may simplify the relationship between input and output of the channel to a single coefficient. Hence, employing a one-tap equalizer may become feasible for signal detection. Furthermore, ZF and MMSE precoding techniques can be applied to the transformed diagonal channel.

[0029] When channel feedback is available at the RAN, a matrix Δ.sup.q,k may be formed from the feedback channel for antenna q from user k by calculating equation (5). Then using the ith diagonal element Δ.sup.q,k[i, i], M×N auxiliary matrixes may be formed as follows:

[00008] $$\tilde{H}_i = [\begin{matrix} {}^{1,1}[i,i] & {}^{1,2}[i,i] & \text{.Math.} & {}^{1,K}[i,i] \\ {}^{2,1}[i,i] & {}^{2,2}[i,i] & \text{.Math.} & {}^{2,K}[i,i] \\ \text{.Math.} & \text{.Math.} & \ddots & \text{.Math.} \\ {}^{N_t,1}[i,i] & {}^{N_t,2}[i,i] & \text{.Math.} & {}^{N_t,K}[i,i] \end{matrix}] \quad 1 \le i \le \mathrm{MN}, \quad (9)$$

where {tilde over (H)}.sup.i is a matrix of the size N.sub.t×K defining a MIMO channel for each diagonal element of the channels. Using {tilde over (H)}.sup.i, beamforming vectors may be formed in the DD domain for each diagonal element one by one. ZF or MMSE may be utilized to do this as follows:

[00009] $${}_i^{\mathrm{MMSE}} = \tilde{H}_i^H (\tilde{H}_i \tilde{H}_i^* + \tfrac{1}{\mathrm{SNR}} I_{N_t \times N_t})^{-1}, \quad (10)$$

$${}_i^{\mathrm{ZF}} = \tilde{H}_i^* (\tilde{H}_i \tilde{H}_i^*)^{-1}, \quad (11)$$

and using Ω.sub.i.sup.ZF the precoder matrices may be generated as follows:

[00010] $$F_{\mathrm{qk}} = {}^{-1} \times \mathrm{diag}({}_1^{\mathrm{ZF}}[q,k], {}_2^{\mathrm{ZF}}[q,k], \text{.Math.} , {}_{\mathrm{NM}}^{\mathrm{ZF}}[q,k]) \times . \quad (12)$$

F.sub.qk may include the same structure as equation (3) and, therefore, may also be block circulant.

[0030] Using equation (2), the received signal at user k after applying the ZF precoder may be provided as:

[00011] $$Y_{\mathrm{DD}}^k = \text{.Math.}_{q=1}^{N_t} H_{\mathrm{DD}}^{q,k} F_{\mathrm{qk}} S_{\mathrm{DD}}^k + Z^k . \quad (13)$$

τ.sub.q=1.sup.N.sup.tH.sub.DD.sup.q,kF.sub.qk=H.sub.eff.sup.k may be a DBC matrix that can be considered as the effective channel between q-th transmit antenna and user-k after applying the precoder. Thus, H.sub.eff.sup.k may be decomposed as H.sub.eff.sup.k=Ξ.sup.−1Δ.sub.eff.sup.kΞ. To estimate the Δ.sub.eff.sup.k at a receiver device, the RAN may transmit preceded pilots for each user. Then the diagonal matrix Δ.sub.eff.sup.kmay be estimated using equation (9). Then, the equalized symbols in the DD domain at user-k may be provided as:

[00012] $$\hat{S}_{\mathrm{DD}}^k = {}^{-1} ({}_{\mathrm{eff}}^k)^{-1} Y_{\mathrm{DD}}^k . \quad (14)$$

[0031] As further shown in FIG. **1**D, and by reference number **115**, the RAN may utilize a ZF and MMSE based precoder for MU-MIMO OTFS in a TF domain for a single-antenna receiver. For example, the RAN may utilize rectangular pulse shaping instead of ideal bi-orthogonal pulse shaping. In this case, the structure of H.sub.DD will be different from equation (8). The method of building up the H.sub.DD matrix for rectangular pulse shaping is described above. Accordingly, H.sub.DD may be provided in an explicit form as:

[00013] $H_{\mathrm{DD}} = \text{.Math.}_{i=1}^{P} h_i (F_N \text{ .Math. } I_M) \;^{l_i} \;^{k_i} (F_N^{*} \text{ .Math. } I_M), \quad (15)$

where Π is a permutation matrix with forward cyclic shift in the form:

[00014] $= \begin{bmatrix} 0 & \text{.Math.} & 0 & 1 \\ 1 & \ddots & \text{.Math.} & 0 \\ \text{.Math.} & \ddots & \ddots & \text{.Math.} \\ 0 & \text{.Math.} & 1 & 0 \end{bmatrix}, \quad (16)$

and Δ=diag{γ.sup.0, γ.sup.1, . . . , γ.sup.MN−1} is a diagonal matrix with

[00015] $\triangleq e^{\frac{j2}{\mathrm{MN}}}$.

Also, l.sub.i and k.sub.i may be related to the delay and Doppler values if each path in the channel as:

[00016] $_i = \frac{l_i}{M \;\; f}, \text{and} \nu_i = \frac{k_i}{\mathrm{NT}}. \quad (17)$

F.sub.N may be defined as an N-point discrete Fourier transform matrix of size N×N and I.sub.M may be ab identity matrix of size M×M and '.Math.' operator may denote the Kronecker product. In some implementations, the RAN may utilize equation (15) to characterize the OTFS channel.

[0032] In some implementations, the RAN may generate MU-MIMO OTFS signals based on ZF and MMSE precoding techniques and where the DBC property of the channel is not available. In such cases, particularly when employing rectangular pulse shaping, the bi-orthogonality property may no longer be guaranteed, resulting in the channel matrix no longer satisfying the DBC property. In some implementations, the RAN may utilize an equivalent TF domain signal to facilitate precoder design in the DD domain.

[0033] In some implementations, the RAN may utilize channel state information at transmitter (CSIT) to calculate the channel response in the TF domain. A single impulse located at a particular location in the DD domain may be transmitted from each antenna of the RAN. Such a signal may be denoted as S.sub.DD[m, n], where m∈{0, . . . , M−1} and n∈{0, . . . , N−1}. A location of the pilot may be selected to be in the middle of the grid to capture the effect of the equivalent channel and avoid the phase rotations happening at the edges of the grid. The transmitted signal in the TF domain may be calculated using the ISFFT of S.sub.DD, as follows:

[00017] $S_{\mathrm{TF}} = \mathrm{ISFFT}(S_{\mathrm{DD}}). \quad (18)$

[0034] The estimated channel from q-th transmit antenna at user-k may be denoted by Ĥ.sub.DD.sup.q,k∈custom-character.sup.MN×MN. Then, the RAN may obtain an estimate of the received signal from q-th transmit antenna at the user-k as follows:

[00018] $\hat{R}_{\mathrm{DD}}^{q,k} = \mathrm{vec}_{M,N}^{-1}(\hat{R}_{\mathrm{DD}}^{q,k}) = \mathrm{vec}_{M,N}^{-1}(\hat{H}_{\mathrm{DD}}^{q,k} S_{\mathrm{DD}}), \quad (19)$

where {circumflex over (R)}.sub.DD.sup.q,k∈custom-character.sup.MN×MN and {circumflex over (R)}.sub.DD.sup.q,k∈custom-character.sup.MN×1. Therefore, the equivalent signal in the TF domain may be:

[00019] $\hat{R}_{\mathrm{TF}}^{q,k} = \mathrm{ISSFT}(\hat{R}_{\mathrm{DD}}^{q,k}) = \mathrm{ISSFT}(\mathrm{vec}_{M,N}^{-1}(\hat{H}_{\mathrm{DD}}^{q,k} S_{\mathrm{DD}})). \quad (20)$

Thus, the transmitted signal and received signal estimate in the TF domain can be calculated at the RAN. By performing element-wise division, an approximate equivalent TF domain channel response may be provided as:

[00020] $\hat{H}_{\mathrm{TF}}^{q,k} = \hat{R}_{\mathrm{TF}} S_{\mathrm{TF}}, \quad (21)$

where 'custom-character' stands for the element-wise division. Ĥ.sub.TF.sup.q,k∈custom-character.sup.M×N may have smaller dimensions compared to the equivalent DD channel. However, the DD domain channel may be sparse, whereas the TF domain channel may not be sparse.

[0035] Now that the channels in the TF domain are known, different precoding techniques may be applied in a similar manner that is applied for OFDM. For each time-frequency bin of a TF resource, a matrix may be generated as follows:

[00021]
$$\tilde{H}_{m,n} = \begin{bmatrix} \hat{H}_{\mathrm{TF}}^{1,1}[m,n] & .\mathrm{Math.} & \hat{H}_{\mathrm{TF}}^{1,K}[m,n] \\ .\mathrm{Math.} & \ddots & .\mathrm{Math.} \\ \hat{H}_{\mathrm{TF}}^{N_t,1}[m,n] & .\mathrm{Math.} & \hat{H}_{\mathrm{TF}}^{N_t,K}[m,n] \end{bmatrix}. \quad (22)$$

[0036] M×N such channel matrices may be formed with a size N.sub.t×K for each matrix. The precoder in the TF domain may be calculated using MMSE and ZF techniques as follows:

[00022]
$$^{\mathrm{MMSE}}_{m,n} = \tilde{H}^{H}_{m,n}\left(\tilde{H}_{m,n}\tilde{H}^{*}_{m,n} + \frac{1}{\mathrm{SNR}} I_{N_t \times N_t}\right)^{-1}, \quad (23)$$

$$^{\mathrm{ZF}}_{m,n} = \tilde{H}^{*}_{m,n}\left(\tilde{H}_{m,n}\tilde{H}^{*}_{m,n}\right)^{-1}, \quad (24)$$

respectively, where Ω.sub.m,n.sup.ZF∈custom-character.sup.N.sup.t.sup.×K or Ω.sub.m,n.sup.MMSE∈custom-character.sup.N.sup.t.sup.×K denote the beamforming coefficients applied to the [m, n] th bin of the data grid of each user in the TF domain which is transmitted from each antenna. For these calculations, the size of the matrix inversion involved may be proportional to the size of the transmitter antennas and, therefore, may be similar to the precoding schemes for OFDM. In order to reduce the calculations further, the beamformers may be calculated with a reduced granularity. By stacking the related beamforming coefficients of each antenna and user in a separate matrix of size M×N, may result in:

[00023]

$$W_{\mathrm{qk}} = \begin{bmatrix} {}_{1,1}[q,k] & .\mathrm{Math.} & {}_{1,N}[q,k] \\ .\mathrm{Math.} & \ddots & .\mathrm{Math.} \\ {}_{M,1}[q,k] & .\mathrm{Math.} & {}_{M,N}[q,k] \end{bmatrix} \quad q \in \{1, .\mathrm{Math.}, N_t\}, k \in \{1, .\mathrm{Math.}, K\}, \quad (25)$$

where the superscripts of "ZF" and "MMSE" have been removed for simplicity. Using equation (25), N.sub.t×K beamforming matrixes may be generated and may be incorporated into the precoding network in the TF domain. To apply this precoder to the signal in the time domain, a Hadamard product of the precoding matrices may be applied to the TF domain signal X.sub.TF, and the signal may be converted to the time domain and transmitted by the RAN.

[0037] In some implementations, the calculated beamforming vectors may be transformed into the DD domain. This may enable the RAN to directly apply the beamforming vectors in the DD domain. The multiplication of the beamforming coefficients in the TF domain may be the same as applying a filter to the signal or equivalently passing the signal through a channel. To calculate the channel response of the calculated precoders in the DD domain, an impulse pilot may be utilized. Therefore, to identify the impulse response of the channel in the DD domain, an impulse pilot may be utilized in the DD domain and the effect of the precoders on the impulse pilot may be calculated.

[0038] The effect of the beamformer matrixes W.sub.qk on the pilot signal in the TF domain may be approximately calculated by Hadamard multiplication of S.sub.TF and W.sub.qk. By applying the SFFT to the result, the impulse response of the beamformers in the DD domain may be calculated as follows:

[00024] $f_{\text{qk}} = \text{SFFT}(S_{\text{TF}} \odot W_{\text{qk}})$. (26)

[0039] In equation (26), f.sub.qk∈custom-character.sup.M×N may be the equivalent DD domain impulse response of the beamforming vector W.sub.qk on the impulse pilot .sub.STF and '⊙' may represent the Hadamard product operation. f.sub.qk,i may be defined as the non-zero elements of f.sub.qk for i∈{0, 1, . . . , V} and α.sub.i and β.sub.i may be the corresponding delay and Doppler bins. Therefore, employing the identical procedure as in equation (15), an equivalent precoding matrix F.sub.qk may be generated from the calculated impulse response f.sub.qk as follows:

[00025] $F_{\text{qk}} \triangleq .\text{Math.}_{i=1}^{V} f_{\text{qk},i}(F_N\ .\text{Math.}\ I_M)\ {}^{i}\ \ {}^{i}(F_N^{*}\ .\text{Math.}\ I_M)$. (27)

[0040] As further shown in FIG. **1**D, and by reference number **120**, the RAN may utilize a block diagonalization (BD) based precoder for MU-MIMO OTFS in a TF domain for a multi-antenna receiver. For example, the methodology described above may be extended to almost all beamforming methods associated with OFDM systems. In some implementations, where users are equipped with N.sub.r antennas, BD precoding may be applied, which is an extension of ZF to the multi-antenna receivers. Assuming the presence of the CSI at the RAN, the TF domain channel matrix of equation (22) may be written as a 3D matrix with {tilde over (H)}.sub.m,n∈custom-character.sup.N.sup.t.sup.×N.sup.r.sup.×K. After calculating the beamforming vectors in the TF, the remaining operations are identical to the operations described above.

[0041] As shown in FIG. **1**E, and by reference number **125**, the RAN may utilize an ISFFT to transform the DD domain symbols into a TF domain signal. For example, the RAN may transform the DD domain symbols X.sub.DD[l′, k′] into a TF domain signal X.sub.TF[m, n] via the ISFFT:

[00026] $X_{\text{TF}}[m,n] = \frac{1}{\sqrt{\text{MN}}}\ .\text{Math.}_{k'=0}^{N-1}\ .\text{Math.}_{l'=0}^{M-1}\ X_{\text{DD}}[l', k']e^{j2\ \left(\frac{nk'}{N} - \frac{ml'}{M}\right)}$, (28)

where m∈{0, . . . , M−1} and n∈{0, . . . , N−1}. At this stage, m may indicate a subcarrier index and n may represent a slot number.

[0042] As shown in FIG. **1**F, and by reference number **130**, the RAN may utilize a Heisenberg transform to convert the TF domain signal to a time domain signal. For example, the RAN may apply the Heisenberg transform to the TF domain signal in order to convert the TF domain signal to the time domain signal. In some implementations, the transformation may include an OFDM transformation, as follows:

[00027] $s(t) = \frac{1}{\sqrt{\text{MN}}}\ .\text{Math.}_{n=0}^{N-1}\ .\text{Math.}_{m=0}^{M-1}\ X_{\text{TF}}[m,n]g_{\text{tx}}(t-\text{NT})e^{j2\ \ m\ \ f(t-\text{NT})}$, (29)

where g.sub.tx(t) is a transmitter pulse shaping filter. Different methods may be utilized to implement the pulse shaping filter, which may impact the performance of the precoding method. In some implementations, the pulse shaping may cause doubly block circulant channel matrices to be generated. In some implementations, a rectangular windowing at the transmitter and receiver side may be utilized. For the rectangular pulse shaping, the g.sub.tx(t) in equation (29) may be defined as:

[00028] $g_{\text{tx}}(t) \triangleq \begin{cases} \frac{1}{\sqrt{T}}, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$. (30)

[0043] As shown in FIG. **1**G, and by reference number **135**, the RAN may add a cyclic prefix to the time domain signal to generate a modified time domain signal. For example, the RAN may add a cyclic prefix to the time domain signal to generate a time domain signal with a cyclic prefix (e.g., referred to herein as the modified time domain signal). A choice of where to add the cyclic prefix may impact the equivalent channel matrix structure in the DD domain. Thus, the quantity of the resources dedicated to the cyclic prefix may be reduced.

[0044] As shown in FIG. **1**H, and by reference number **140**, the UE may convert the modified time domain signal to the TF domain signal and may convert the TF domain signal to the DD domain symbols. For example, the RAN may transmit the modified time domain signal to one of the UEs, and the UE may receive the modified time domain signal. At the receiver (e.g., the UE), the reverse procedure may be applied. First the UE may apply a windowing function to the modified time domain signal and may utilize a Wigner transform to convert the modified time domain signal to the TF domain signal. The UE may utilize an SFFT to convert the TF domain signal back to the DD domain symbols. The UE may perform equalization and signal detection based on the DD domain symbols.

[0045] In this way, the RAN applies ZF and MMSE to MU-MIMO OTFS networks. For example, a RAN may apply ZF and/or MMSE models to beamforming techniques for MU-MIMO OTFS systems. Although the models are described in connection with MU-MIMO, the models may also be used in SU-MIMO OTFS. In a ZF precoding scheme for OTFS, a precoder for each antenna and each user may be designed to cancel out total interference to the other users. An MMSE precoder for OTFS may incorporate benefits of MRT and ZF precoding schemes considering both signal and interference levels to achieve a robust performance. A transformation may be utilized to diagonalize an effective channel matrix of the system, and a precoding design may be performed accordingly. Alternatively, beamformers may be calculated and applied in the TF domain rather than the DD domain. Thus, the RAN conserves computing resources, networking resources, and/or the like that would otherwise have been consumed by increasing power consumption, reducing battery life, handling inter-carrier interference, reducing spectral efficiency, generating a degraded quality signal, and/or the like.

[0046] As indicated above, FIGS. **1**A-**1**H are provided as an example. Other examples may differ from what is described with regard to FIGS. **1**A-**1**H. The number and arrangement of devices shown in FIGS. **1**A-**1**H are provided as an example. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIGS. **1**A-**1**H. Furthermore, two or more devices shown in FIGS. **1**A-**1**H may be implemented within a single device, or a single device shown in FIGS. **1**A-**1**H may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) shown in FIGS. **1**A-**1**H may perform one or more functions described as being performed by another set of devices shown in FIGS. **1**A-**1**H.

[0047] FIG. **2** is a diagram of an example environment **200** in which systems and/or methods described herein may be implemented. As shown in FIG. **2**, the environment **200** may include a UE **210**, a RAN **220**, and/or a core network **230**. Devices and/or elements of the environment **200** may interconnect via wired connections and/or wireless connections.

[0048] The UE **210** includes one or more devices capable of receiving, generating, storing, processing, and/or providing information, such as information described herein. For example, the UE **210** can include a mobile phone (e.g., a smart phone or a radiotelephone), a laptop computer, a tablet computer, a desktop computer, a handheld computer, a gaming device, a wearable communication device (e.g., a smart watch or a pair of smart glasses), a mobile hotspot device, a fixed wireless access device, customer premises equipment, an autonomous vehicle, or a similar type of device.

[0049] The RAN **220** may support, for example, a cellular radio access technology (RAT). The RAN **220** may include one or more base stations (e.g., base transceiver stations, radio base stations, node Bs, eNodeBs (eNBs), gNodeBs (gNBs), base station subsystems, cellular sites, cellular towers, access points, transmit receive points (TRPs), radio access nodes, macrocell base stations, microcell base stations, picocell base stations, femtocell base stations, or similar types of devices) and other network entities that can support wireless communication for the UE **210**. The RAN **220** may transfer traffic between the UE **210** (e.g., using a cellular RAT), one or more base stations (e.g., using a wireless interface or a backhaul interface, such as a wired backhaul interface), and/or the core network **230**. The RAN **220** may provide one or more cells that cover geographic areas.

[0050] In some implementations, the RAN **220** may perform scheduling and/or resource management for the UE **210** covered by the RAN **220** (e.g., the UE **210** covered by a cell provided by the RAN **220**). In some implementations, the RAN **220** may be controlled or coordinated by a network controller, which may perform load balancing, network-level configuration, and/or other operations. The network controller may communicate with the RAN **220** via a wireless or wireline backhaul. In some implementations, the RAN **220** may include a network controller, a self-organizing network (SON) module or component, or a similar module or component. In other words, the RAN **220** may perform network control, scheduling, and/or network management functions (e.g., for uplink, downlink, and/or sidelink communications of the UE **210** covered by the RAN **220**).

[0051] In some implementations, the core network **230** may include an example functional architecture in which systems and/or methods described herein may be implemented. For example, the core network **230** may include an example architecture of a fifth generation (5G) next generation (NG) core network included in a 5G wireless telecommunications system, a sixth generation (6G) core network included in a 6G wireless telecommunications system, and/or the like. While the example architecture of the core network **230** shown in FIG. **2** may be an example of a service-based architecture, in some implementations, the core network **230** may be implemented as a reference-point architecture and/or a fourth generation (4G) core network, among other examples.

[0052] The number and arrangement of devices and networks shown in FIG. **2** are provided as an example. In practice, there may be additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than those shown in FIG. **2**. Furthermore, two or more devices shown in FIG. **2** may be implemented within a single device, or a single device shown in FIG. **2** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of the environment **200** may perform one or more functions described as being performed by another set of devices of the environment **200**.

[0053] FIG. **3** is a diagram of example components of a device **300**, which may correspond to the UE **210** and/or the RAN **220**. In some implementations, the UE **210** and/or the RAN **220** may include one or more devices **300** and/or one or more components of the device **300**. As shown in FIG. **3**, the device **300** may include a bus **310**, a processor **320**, a memory **330**, an input component **340**, an output component **350**, and a communication component **360**.

[0054] The bus **310** includes one or more components that enable wired and/or wireless communication among the components of the device **300**. The bus **310** may couple together two or more components of FIG. **3**, such as via operative coupling, communicative coupling, electronic coupling, and/or electric coupling. The processor **320** includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. The processor **320** is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, the processor **320** includes one or more processors capable of being programmed to perform one or more operations or processes described elsewhere herein.

[0055] The memory **330** includes volatile and/or nonvolatile memory. For example, the memory **330** may include random access memory (RAM), read only memory (ROM), a hard disk drive, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory). The memory **330** may include internal memory (e.g., RAM, ROM, or a hard disk drive) and/or removable memory (e.g., removable via a universal serial bus connection). The memory **330** may be a non-transitory computer-readable medium. The memory **330** stores information, instructions, and/or software (e.g., one or more software applications) related to the operation of the device **300**. In some implementations, the memory **330** includes one or more memories that are coupled to one or more processors (e.g., the processor **320**), such as via the bus **310**.

[0056] The input component **340** enables the device **300** to receive input, such as user input and/or sensed input. For example, the input component **340** may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system sensor, an accelerometer, a gyroscope, and/or an actuator. The output component **350** enables the device **300** to provide output, such as via a display, a speaker, and/or a light-emitting diode. The communication component **360** enables the device **300** to communicate with other devices via a wired connection and/or a wireless connection. For example, the communication component **360** may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

[0057] The device **300** may perform one or more operations or processes described herein. For example, a non-transitory computer-readable medium (e.g., the memory **330**) may store a set of instructions (e.g., one or more instructions or code) for execution by the processor **320**. The processor **320** may execute the set of instructions to perform one or more operations or processes described herein. In some implementations, execution of the set of instructions, by one or more processors **320**, causes the one or more processors **320** and/or the device **300** to perform one or more operations or processes described herein. In some implementations, hardwired circuitry may be used instead of or in combination with the instructions to perform one or more operations or processes described herein. Additionally, or alternatively, the processor **320** may be configured to perform one or more operations or processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0058] The number and arrangement of components shown in FIG. **3** are provided as an example. The device **300** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **3**. Additionally, or alternatively, a set of components (e.g., one or more components) of the device **300** may perform one or more functions described as being performed by another set of components of the device **300**.

[0059] FIG. **4** is a flowchart of an example process **400** for applying ZF and MMSE to MU-MIMO OTFS networks. In some implementations, one or more process blocks of FIG. **4** may be performed by a device (e.g., the RAN **220**). In some implementations, one or more process blocks of FIG. **4** may be performed by another device or a group of devices separate from or including the device, such as a UE (e.g., the UE **210**). Additionally, or alternatively, one or more process blocks of FIG. **4** may be performed by one or more components of the device **300**, such as the processor **320**, the memory **330**, the input component **340**, the output component **350**, and/or the communication component **360**.

[0060] As shown in FIG. **4**, process **400** may include applying a precoder to input information symbols to generate delay-Doppler (DD) domain symbols (block **410**). For example, the device may apply a precoder to input information symbols to generate delay-Doppler (DD) domain symbols, as described above. In some implementations, the precoder is a zero forcing and minimum mean square error based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a DD domain based on a doubly block circulant property of a channel matrix. In some implementations, the precoder is a zero forcing and minimum mean square error based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a time-frequency domain for a single-antenna receiver. In some implementations, process **400** includes converting the zero forcing and minimum mean square error based precoder to a DD domain.

[0061] In some implementations, the precoder is a block diagonalization based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a time-frequency domain for a multi-antenna receiver. In some implementations, process **400** includes converting the block diagonalization based precoder to a DD domain.

[0062] In some implementations, the device is a radio access network. In some implementations, the precoder provides a zero forcing precoding scheme. In some implementations, the precoder provides a minimum mean square error precoding scheme. In some implementations, the minimum mean square error precoding scheme includes a zero forcing precoding scheme and a maximum ratio transmission precoding scheme. In some implementations, the device is a multi-user multi-input multi-output (MIMO) orthogonal time and frequency space (OTFS) system or a single-user MIMO OTFS system.

[0063] As further shown in FIG. **4**, process **400** may include utilizing an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal (block **420**). For example, the device may utilize an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal, as described above.

[0064] As further shown in FIG. **4**, process **400** may include utilizing a Heisenberg transform to convert the TF domain signal to a time domain signal (block **430**). For example, the device may utilize a Heisenberg transform to convert the TF domain signal to a time domain signal, as described above. In some implementations, the inverse symplectic finite Fourier transform and the Heisenberg transform are provided in an orthogonal time and frequency space modulator of the device.

[0065] As further shown in FIG. **4**, process **400** may include adding a cyclic prefix to the time domain signal to generate a modified time domain signal (block **440**). For example, the device may add a cyclic prefix to the time domain signal to generate a modified time domain signal, as described above.

[0066] As further shown in FIG. **4**, process **400** may include transmitting the modified time domain signal to at least one user equipment (block **450**). For example, the device may transmit the modified time domain signal to at least one UE, as described above. In some implementations, transmitting the modified time domain signal causes the at least one UE to convert the modified time domain signal to the TF domain signal and to convert the TF domain signal to the DD domain symbols.

[0067] Although FIG. **4** shows example blocks of process **400**, in some implementations, process **400** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **4**. Additionally, or alternatively, two or more of the blocks of process **400** may be performed in parallel.

[0068] The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the implementations to the precise form disclosed. Modifications may be made in light of the above disclosure or may be acquired from practice of the implementations.

[0069] As used herein, the term “component” is intended to be broadly construed as hardware, firmware, or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware, firmware, and/or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the implementations. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code-it being understood that software and hardware can be used to implement the systems and/or methods based on the description herein.

[0070] As used herein, satisfying a threshold may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, and/or the like, depending on the context.

[0071] Although particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one claim, the disclosure of various implementations includes each dependent claim in combination with every other claim in the claim set.

[0072] No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include

one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the term “set” is intended to include one or more items (e.g., related items, unrelated items, a combination of related and unrelated items, and/or the like), and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).
[0073] In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Claims

1. A method, comprising: applying, by a device, a precoder to input information symbols to generate delay-Doppler (DD) domain symbols; utilizing, by the device, an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal; utilizing, by the device, a Heisenberg transform to convert the TF domain signal to a time domain signal; adding, by the device, a cyclic prefix to the time domain signal to generate a modified time domain signal; and transmitting, by the device, the modified time domain signal to at least one user equipment.
2. The method of claim 1, wherein transmitting the modified time domain signal causes the at least one user equipment to convert the modified time domain signal to the TF domain signal and to convert the TF domain signal to the DD domain symbols.
3. The method of claim 1, wherein the precoder is a zero forcing and minimum mean square error based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a DD domain based on a doubly block circulant property of a channel matrix.
4. The method of claim 1, wherein the precoder is a zero forcing and minimum mean square error based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a time-frequency domain for a single-antenna receiver.
5. The method of claim 4, further comprising: converting the zero forcing and minimum mean square error based precoder to a DD domain.
6. The method of claim 1, wherein the precoder is a block diagonalization based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a time-frequency domain for a multi-antenna receiver.
7. The method of claim 6, further comprising: converting the block diagonalization based precoder to a DD domain.
8. A device, comprising: one or more memories; and one or more processors, coupled to the one or more memories, configured to: apply a precoder to input information symbols to generate delay-Doppler (DD) domain symbols; utilize an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal; utilize a Heisenberg transform to convert the TF domain signal to a time domain signal; add a cyclic prefix to the time domain signal to generate a modified time domain signal; and transmit the modified time domain signal to at least one user equipment to cause the at least one user equipment to convert the modified time

domain signal to the TF domain signal and to convert the TF domain signal to the DD domain symbols.
9. The device of claim 8, wherein the inverse symplectic finite Fourier transform and the Heisenberg transform are provided in an orthogonal time and frequency space modulator of the device.
10. The device of claim 8, wherein the device is a radio access network.
11. The device of claim 8, wherein the precoder provides a zero forcing precoding scheme.
12. The device of claim 8, wherein the precoder provides a minimum mean square error precoding scheme.
13. The device of claim 12, wherein the minimum mean square error precoding scheme includes a zero forcing precoding scheme and a maximum ratio transmission precoding scheme.
14. The device of claim 8, wherein the device is a multi-user multi-input multi-output (MIMO) orthogonal time and frequency space (OTFS) system or a single-user MIMO OTFS system.
15. A non-transitory computer-readable medium storing a set of instructions, the set of instructions comprising: one or more instructions that, when executed by one or more processors of a device, cause the device to: apply a precoder to input information symbols to generate delay-Doppler (DD) domain symbols; utilize an inverse symplectic finite Fourier transform to transform the DD domain symbols into a time-frequency (TF) domain signal; utilize a Heisenberg transform to convert the TF domain signal to a time domain signal, wherein the inverse symplectic finite Fourier transform and the Heisenberg transform are provided in an orthogonal time and frequency space modulator of the device; add a cyclic prefix to the time domain signal to generate a modified time domain signal; and transmit the modified time domain signal to at least one user equipment.
16. The non-transitory computer-readable medium of claim 15, wherein the precoder is a zero forcing and minimum mean square error based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a DD domain based on a doubly block circulant property of a channel matrix.
17. The non-transitory computer-readable medium of claim 15, wherein the precoder is a zero forcing and minimum mean square error based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a time-frequency domain for a single-antenna receiver.
18. The non-transitory computer-readable medium of claim 17, wherein the one or more instructions further cause the device to: convert the zero forcing and minimum mean square error based precoder to a DD domain.
19. The non-transitory computer-readable medium of claim 15, wherein the precoder is a block diagonalization based precoder for multi-user multi-input multi-output orthogonal time and frequency space in a time-frequency domain for a multi-antenna receiver.
20. The non-transitory computer-readable medium of claim 19, wherein the one or more instructions further cause the device to: convert the block diagonalization based precoder to a DD domain.